2024 International Conference on Trends in Quantum Computing and Emerging Business Technologies
CHRIST (Deemed to be University), Pune Lavasa Campus, India. Mar 22-23, 2024

2024 International Conference on Trends in Quantum Computing and Emerging Business Technologies (TQCEBT) | 979-8-3503-8427-7/24/$31.00 ©2024 IEEE | DOI: 10.1109/TQCEBT59414.2024.10545140

Customer Lifetime Value Prediction: An In-Depth Exploration with Regression, Regularization and Hyperparameter Tuning

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***Abstract*:-In today's dynamic business environment, companies have been strategically shifting towards a customer-centric approach from their traditional product-centric focus. The main goal of this paper is to estimate customer lifetime value of 5,000 customers in the retail industry. This research follows a step-by-step approach to construct a multiple regression machine learning model. The model used in the study is based on the nine features to predict the customer life time value. First basic train-test split model is developed, which predicted 74% of variation in the customer lifetime value. This necessitates to improve the model performance, hence to address the multicollinearity problem lasso regularization is used. After lasso regularization , final model is trained with hyperparameter turning for further model performance improvement. The results show significant improvements in predicting customer lifetime value with the final model. This study suggests that the machine learning regression models can help to businesses to better understand how much value they can generate from individual customer.This deep understanding about customers helps retail businesses to align their customer engagement strategies to create a positive impact on the profitability and maximizing overall value offered to the customers.**

Keywords: - Customer life time value prediction, Regularization, Hyperparameter turning, Learning curve

I. INTRODUCTION

Companies have been changing their approach in business strategies from product- oriented to customer oriented. [1]The importance of customer lifetime value has a received a lot of attention in academic research and business practices. The companies who are aware about the factors influencing customer lifetime value are highly interested its measurements.[2]Customer lifetime value analysis helps to gain key insights about individual customer behaviour and preferences which helps these companies in customer's segmentation. [3]Individual customer value analysis helpsevery business to identify its most valued customer and to differentiate less valuable customers to optimize marketing campaigns.Companies by understanding the churn rate take timely actions to retain the customers by offering incentives, targeted events, free gifts.According to [4], they identified a linear correlation between profit per customer and customer lifetime value, as well as an exponential relationship between retention rate and customer lifetime value. Therefore, it is recommended that companies prioritize business strategies that enhance customer retention rates for optimal results.

This paper employed a regression machine learning model to predict customer lifetime value, leveraging data from the retail industry. The primary aim of this study is twofold: firstly, to forecast customer lifetime value, and secondly, to explore methods for enhancing the overall performance of the predictive model. The main thrust of this research is on utilizing regression techniques to recognize patterns in retail data, extracting insights that pertain to customer lifetime value

II. RELATED WORKS

[4] analysed the customer life time value with the real time data of online shopping platform with the objective of customer segmentation and value calculation. The study found that the model accuracy of 90%,91%,91%,87% for K-nearest neighbours, logistic regression, support vector machine, decision tree respectively. Based on historical transactionconsumer’s survival probability calculated and it is found that as consumer purchases retention probability increases and the same gradually decreases over the period. [5] based on the historical transaction customer life time value is calculated by using UCI Machine Learning Repository’s Online Realtime Dataset. The study found that the customer has tendency of repeat purchasing with 12 to 50 days or twelve purchases in the year. The linear regression outcome shows that r-square value is 71%. This study findings can help the businesses to develop business strategies and marketing strategies along with the marketing budget to acquire new customers. [6] explained businesses strategies have beenshifting from being product centric to customer centric and focusing on improving customer experience. This study used to predict customer life timevaluewith the Gradient Boosting Regressor model and model predicted 79%,84% $R^2$for linear regression, gradient boosting regressor respectively. The study's customer segmentation analysis reveals that a specific subset, comprising 17% of business customers, is responsible for 50% of the overall business's contribution. The company needs to concentrate its efforts to boost overall business

profitability by focusing this customer segment.[7] analysed customer lifetime value along with two crucial factors responsible for business revenue i.e. distinct product category and trend in amount spent. Researcher employed deep neural network with multi output, multi output random forest and multi output decision tree for prediction.The study found that the multi output deep neural network outperformed relatively compared to the other two models. Managerial implications of the study that it is easy and less costly to adopt only one model for online shopping platforms. [8] tested customer life time value in the banking industry with the machine learning techniques with the arbitrary time horizons and product-based propensity models. The noteworthy findings is that the top 10% of customers exhibited a 3.2-fold higher likelihood for purchasing an investment product in the subsequent year. [9] examined the data of 8099 samples to determine the customer lifetime value with the linear regression, support vector machine, neural network, random forest. The result of study indicates that monthly autopremium, total claim amount, coverage are key factors influencing customer life value. Another noteworthy finding is out of these four machine learning models random forest performed best.

[10] evaluated various machine learning techniques with the hierarchical ensembled prediction framework for the sample data of B2B companies. The study measured model efficacy with the multiple metric and found improvement in the model performance as compared traditional measurement models. [11] in their study developed model with the novel approach by incorporating temporary customer purchasing behaviour. Researcher followed multiple machine learning techniques but at its core used deep learning network. [12]analysed the customers with the unsupervised machine learning algorithm cluster analysis to cluster the customers by self-organizing map. This innovative approach can better resolve the manager's problem to devising marketing strategies by understanding their customer's better. [13] discussed the customer segmentation findings based on the customer life time value model with neural network technique by using 56000 customers data of Taavon Bank. Researchers first found optimum numbers of clusters in the data and then applied Kohonen neural network. Based on the study findings out in the first , second phase of the results 33%, 50% customers were chosen and in the third phase 16.50% of total customers were chosen as golden customers for the bank. Now the bank manager can design marketing strategies to address the 10,986 customers out of total 56,000 customers. [14] studied the customer's behaviour to predict the customer life time value and for the analysis textile company's sales data is considered. Researchers compared their findings of stacked machine learning methods with the deep neural network, bagging support vector regression, light gradient boosting techniques. The study found that the stack machine learningmethod was superior as compared to the other methods. [15]analysed customer life time value prediction with the regression modelling. This study included 24 variables however with the help of feature engineering the same is reduced to 14 variables. This study is based on the linear regressor, k-nearest neighbour regression, decision tree regressor, gradient boosting regressor. The study findings indicates that the gradient boosting regressor is best performer as compared to the other models. This study recommends that company need to alter its business strategy by attracting luxury car owners to enhance the customer life time value segmentation. [16] applied various machine learning models on non-contractual freemium users' data for the prediction of individual level customer life time value. This dataset was highly imbalanced as only few users are paying for services and hence to avoid this problem researchers used synthetic minatory oversampling (SMOTE). The study findings suggests that the SMOTE helped in the better prediction performance and to design business strategies around premium and high-value users. [17] deployed multiple machine learning models to predict customer segment to improve the customer retention in online retail business. This study includes logistic regression, XGB classifier, K Neighbours classifier, Decision tree, random forest classifier, AdaBoost classifier, SVC. The result of the study suggests that the decision tree and XGB classifier found superior performer as compared to other models. Out of these two models researchers move ahead with XGB classifier for further investigation with hyperparameter tuning and done more feature engineering to improve the model performance. [18] analysed the customer life time value through global super store dataset with 10,000 plus transactions. The model was trained with random forest algorithm and which followed by random search turning to attain the superior accuracy. The study findings indicate that random search best hyperparameter technique suited best in classifying the dataset as compared to the random forest, AdaBoost model.

III. PROPOSED WORK

This study focuses on predicting customer lifetime value in the retail industry with 5000 customers dataset which include 9 features. This study used synthetic dataset to predict customer lifetime value. Table 1 provides a suitable explanation of features and target variable used in the study. The data use in this study is synthetic in nature.

TABLE I. VARIABLE DEFINITION

Variable name	Definition
Purchase Frequency	Number of purchases made by the customer
Average Purchase Value	Average amount spent by the customer in each transaction
Customer Satisfaction Score	Rating indicating the level of satisfaction
Customer Engagement	Level of engagement categorized as Low, Moderate, or High
Customer Acquisition Cost	Cost associated with acquiring a new customer
Retention Rate	Percentage of customers retained over a specific period
Time Since Last Purchase Months	Duration since the customer's last transaction
Age	Age of the customer
Churn Rate	Percentage of customers lost over a given period
Customer Lifetime Value	Estimated total revenue from a customer throughout their entire relationship

The study followed systematic process to investigate the influence of features on customer lifetime value. Initially basic train-test split model is developed to understand the model ability to predict the target variable. These basic steps helped to evaluate the model performance and sets the stage for the model improvement. To address the overfitting problem of model and to enhance the model generalization, Lasso regularization is used. Lasso imposes a penalty on less or non-performing features and prevent them from dominating the model, hence the ability of model enhances to correctly predict to unseen data.In last hyperparameter tuning is used to enhance the predictive performance of the model. This is iterative process with multiple hyperparameter configurations with the objective to select those values that helps to enhance the model's performance and generalizing ability. The objective of hyperparameter tuning is to prevent the overfitting of model on the training data and building a strong model to predict customer lifetime value.Learning curve visualisation is used to evaluate the predictive accuracy by plotting training and validation performance against varying dataset sizes. By visually analysing the convergence or divergence insights gained about model's ability to generalize across different data sizes.

IV. MODEL EXPERIMENTATION

This section provides detailed discussion about the regression model results and performances with basic train-test split model, Lasso regularization model and hyperparameter tuning model.

Table 2 presents the results of basic train-test split regression model. The findings suggests that the model is trained with 4000 customers data size i.e. 80% of data and tested on 1000 customers data i.e. 20% of data. The trained and test model has ability to predict 74.21% and 74.36% of variation in customer lifetime value.

A. *Basic train-test split model*

TABLE II. TRAIN-TEST SPLIT

Model results	Training model	Test model
Data size	4000	1000
Data allocation	0.20	0.80
R^2	0.7421	0.7436

B. *Learning curve for train-test split*

As per the results training model R^2 value 74.21% and test model is 74.36% that indicates the model is not facing underfitting or overfitting problem. The model is converging and stabilizing at 74% level so there is no need for further training.

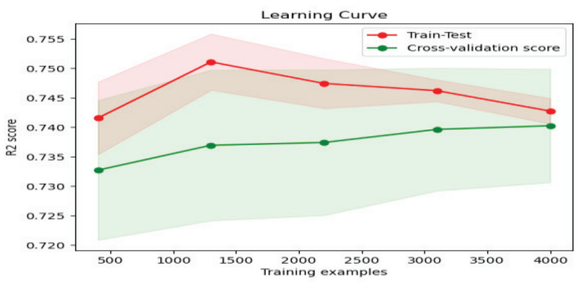

Fig. 1. Learning curve of train-test split model

C. *Lasso regularization*

This model is built-up with 9 features which is resulting in multicollinearity problem and affecting adversely to model performance. Hence to address this problem and to improve model performance lasso regularization is tested with alpha values 0.01,0.1,1 and1.0. The findings reported in table 3 suggests that the best alpha value is 0.01 and relatively low level of regulation is required for optimal model performance. The findings of mean cross-validation $R^2$is 94%that can be interpreted as across different subset of data on average 94% variance is explained by model. This finding confirms that the model performance is improved as compared to the basic train-test split model.

TABLE III. REGULARIZATION

Model parameters	Output
Best Alpha(Lasso)	0.01
Mean Cross-Validation R^2	0.94
R^2 value in training dataset	0.9511
R^2 value in test dataset	0.9488

D. *Learning curve for lasso regularization*

Plot 2 present learning curve for lasso regularization. Learning curve is showing steady growth in R^2 value of training and test model with 95.11% and 94.88 respectively. After the training set hits 3000 marks both the model convergence is visible. That suggests that the further addition of data will not change the output significantly.

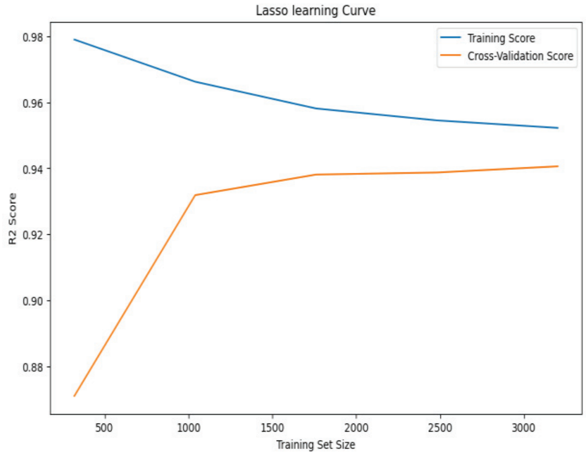

Fig. 2. Lasso regularization model learning curve

E. *Hyperparameter tuning*

Table 4 presents the results of hyperparameter tuning model. To test the model performance, model ran on polynomial degree of 2,3,4 and5. The findings suggests that the out of these 2,3,4,5 polynomial degree 4 is the best degree found during hyperparameter tuning process to improve model performance. Table 4 findings suggests that the mean cross-validation R^2 is94.81% that indicates the polynomial regression model predict 94.81% variation in customer lifetime value across different subset of data. The performance of R^2 training and test dataset is 96.30 and 96.87 respectively which is found highest when compared with the basic train-test split model, lasso regression model.

TABLE IV. HYPERPARAMETER TUNING

Model parameters	Output
Best Degree (Hyperparameter Tuning)	4
Mean Cross-Validation R^2 (Best Model)	0.9481
R^2 Score on Training Set	0.9630
R^2 Score on Test Set	0.9687

F. Hyperparameter tuning learning curve

Plot 3 presents learning curve for hyperparameter tuning learning curve. Training curve's consistent line close to 96% showmodel's proficiency in capturing patterns within the training data. The straight trajectory of the blue line suggests that additional training instances beyond the initial 1500 data points do not substantially improve performance, indicating a plateau in learning. The convergence indicates that the model's performance on the validation set aligns closely with its performance on the training set. The proximity of the blue and orange lines underscores minimal overfitting or underfitting.

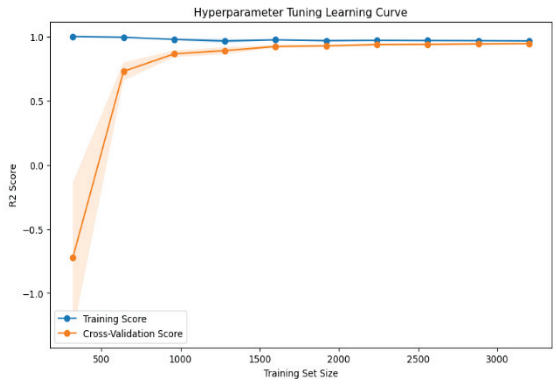

Fig. 3. Hyperparameter tuning learning curve

V. CONCLUSION AND FUTURE WORK

This study focuses on experimenting with regression models to predict customer lifetime value in the retail industry. The initial model employs a basic train-test split, revealing a predictive ability of 74.21% and 74.36% for training and test datasets, respectively. However, to address the problem of multicollinearity due to the 9 features necessitates the application of lasso regularization. The result of lasso regularization model indicates optimal performance with an alpha value of 0.01. The mean cross-validation R^2 for lasso regularization is 94% that indicates substantial improvement over the basic model. To improve further model performance hyperparameter tuning is used with a polynomial degree of 4. The tuned model demonstrates a mean cross-validation R^2 of 94.81%, and R^2 scores of 96.30% and 96.87% for training and test datasets, respectively. Overall, these findings suggest that both lasso regularization and hyperparameter tuning significantly enhance the predictive performance of the regression model in predicting customer lifetime value.

A. Scope for future work

In future the customer lifetime value can be predicted with the different key features of retail industry affecting to the customer lifetime value. Apart from these features specialized algorithms need to be explored to improve the model performance like ridge regression, elastic net regression, huber regression, generalized addictive models and principles components regression. The retail industry is very dynamic one about additional feature information can be extracted with social media trends can be additional aid to improve the model's predictive ability.

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